

Program: Diploma In Electrical Engineering	
Course Code: 5549A	Course Title: Power Electronics Laboratory
Semester: 5	Credits: 1.5
Course Category: Program Elective	
Periods per week: 3 (L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- To familiarize with various applications of Op-Amps
- To identify the power electronics devices and their circuits in electrical engineering.
- To know about industrial applications of power electronics

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Basic semi conductor device operation	2039	Fundamentals of Electrical & Electronics Lab	2
Transistors and OP-amps	3031	Analog & Digital circuits	3
Basics of electronic circuits	4036	Electronics Lab	4
Power semiconductor devices	4651	PowerElectronics	4

Course Outcomes:

On completion of the course, the students will be able to:

CO_n	Description	Duration (Hours)	Cognitive level
CO1	Develop various op-amp circuits using IC741C	9	Applying
CO2	Identify various power semiconductor devices and plot the V-I characteristics of SCR	6	Applying
CO3	Build controlled rectifier circuits using SCR and set up a triac firing circuit.	9	Applying
CO4	Construct various industrial control circuits using power semi Conductor devices.	15	Applying
	Lab Exam	6	

CO - PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1				3			
CO2				3			
CO3				3			
CO4				3			

3- Strongly mapped ,2- Moderately mapped ,1- Weakly mapped

Course Out line

Module Outcomes	Name of Experiment	Duration (Hours)	Cognitive Level
CO1	Develop various op-amp circuits using IC741C		
M1.01	Adder circuit	1.5	Applying
M1.02	Integrator	1.5	Applying
M1.03	Differentiator	1.5	Applying
M1.04	Comparator	1.5	Applying
M1.05	Zero Crossing Detector	1.5	Applying
M1.06	Schmitt trigger	1.5	Applying
CO2	Identify various power semiconductor devices and plot the V-I characteristics of SCR		
M2.01	Identify various terminals and check the following power semiconductor devices (a) SCR (b) DIAC (c) TRIAC (d) UJT (e) IGBT	1	Applying
	MOSFET		
M2.02	Set up a UJT relaxation oscillator circuit and observe the waveform	2	Applying
M2.03	Plot the VI characteristics of SCR	3	Applying
	Lab Exam	3	
CO3	Build controlled rectifier circuits using SCR and setup a triac firing circuit.		
M3.01	Construct a single phase half wave controlled rectifier circuit using SCR for resistive load and observe the wave forms a cross load and SCR.	3	Applying
M3.02	Construct a single phase half controlled rectifier circuit using SCR with R load and observe the	3	Applying

	wave forms across load and SCR.		
M3.03	Setup a TRIAC Firing circuit Using DIAC and observe the wave forms across TRIAC and load	3	Applying
CO4:	Construct various industrial control circuits using power semiconductor devices.		
M4.01	Setup a DC motor speed control circuit using SCR and verify the output	3	Applying
M4.02	Setup an emergency lamp circuit using SCR	3	Applying
M4.03	Setup a of battery charger circuit using thyristors.	3	Applying
M4.04	Assemble a fan regulator using thyristors	3	Applying
M4.05	Construct an automatic street lamp circuit using LDR	3	Applying
	Lab Exam	3	

Text / Reference:

T/R	Book Title / Author
T1	K.A Navas , Electronics Lab Manual – Vol 1 PHI Learning , ISBN : 9788120351424
T2	K.A Navas, Electronics Lab Manual – Vol 2 , PHI Learning , ISBN : 9789388028080
R1	Rama moorthy M ., An Introduction to Thyristors and their applications , East – West Press Pvt. Ltd., New Delhi , ISBN : 8185336679
R2	Sugandhi , Rajendra Kumar and Sugandhi , Krishna Kumar , Thyristors : Theory and Applications , New Age Internation l (P) ltd. Publishers , NewDelhi , ISBN: 978-0-85226-852-0.
R3	Bhattacharya , S.K., Fundamentals of Power Electronics , Vikas Publishing House Pvt. Ltd.Noida. ISBN : 978-8125918530
R4	Zbar , PaulB., Industrial Electronics : A Text – Lab Manual ,Mc Graw Hill

	Publishing Co. Ltd. , New Delhi , ISBN : 978-0070728226.
R5	Singh , M.D. and Khanchandani ,K.B. , Power Electronics, Tata Mc Graw Hill Publishing Co. Ltd , New Delhi , 2008 ISBN : 9780070583894.
R6	Grafham D.R., SCR Manual , General Electric Co. , ISBN : 978-0137967711.
R7	Jain & Alok , Power Electronics and its Applications , Penram International Publishing (India) Pvt. Ltd , Mumbai , ISBN : 978-8187972228.
R8	Rashid , Muhammad , Power Electronics Circuits Devices and Applications, Pears on Education India , Noida , ISBN : 978-0133125900.

Online Resources:

Sl No	Website Link
1	www.circuitstoday.com
2	wikipedia.org
3	>Courses>>ElectricalEngineering">NPTEL>>Courses>>ElectricalEngineering
4	www.youtube.com
5	http://www.electronicsteacher.com
6	https://www.all about circuits.com/textbook/experiments

Student Activity

Suggested Open – ended Experiments:

Students can do open ended experiments as a group of 3-5. There is no duplication in experiments in between groups. This is mainly for the purpose of continuous internal evaluation Students should prepare a separate report on open ended experiment of their choice.

Example : Design and develop power electronic circuits for various domestic applications.

Design a rectifier circuit using Op-Amps